



# KALPESH ASHVINBHAI VALA

B.Tech. - Computer Science & Engineering

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## BRIEF SUMMARY

Passionate and self-driven Computer Science Engineering student with a strong foundation in Java, Python, C++, JavaScript, and Go. Experienced in developing full-stack software and web applications, with a focus on backend development and scalable system design. Skilled in Spring Boot, Flask, Express.js, Next.js, Node.js, and web technologies like ReactJS and Tailwind CSS. Currently exploring Kubernetes and the Spring framework. Demonstrates a hands-on approach to building efficient, real-world solutions, including a real-time chat application and an e-commerce backend system. Actively involved in projects and hackathons, with a consistent drive to learn and experiment with new technologies.

## KEY EXPERTISE

Java ReactJS Docker Kubernetes Go Javascript Python HTML/CSS Spring Boot Flask Tailwind CSS  
Gin MySQL PostgreSQL MongoDB Redis Git JWT WebSockets Linux AWS

## EDUCATION

### Institute of technology, Nirma University

2022 - 2026

B.Tech. - Computer Science & Engineering | CGPA: 8.16 / 10

### Nutan Vidhyapith, Mahuva

2022

12<sup>th</sup> | GSEB | Percentage: 88.31 / 100

### Nutan Vidhyapith, Mahuva

2020

10<sup>th</sup> | GSEB | Percentage: 87.67 / 100

Minor Subject : Cyber Physical System

## AWARDS AND SCHOLARSHIPS

- Vidyadhan Scholar – Sarojini Damodaran Foundation (SDF) Awarded a merit-based scholarship for academic excellence and commitment to education; recognized among top-performing students in Gujarat.

## INTERNSHIPS

### The Special Character PVT. LTD. | IT / Computers - Software

08 May, 2025 - 07 Jul, 2025

Full Stack Developer

Key Skills: Github Docker Medusa.js AWS Deployment Git Payload CMS Task Management

Fullstack Developer

- Spearheaded the end-to-end development of a full-fledged e-commerce platform, leveraging Medusa.js for core commerce logic and Payload CMS for dynamic content management.
- Engineered key backend modules, including secure user authentication, comprehensive product lifecycle management, and custom CMS workflows to streamline administrative operations.
- Established and enforced a Git-based workflow, mandating pull request (PR) reviews to maintain high code quality, ensure branch integrity, and resolve merge conflicts effectively.
- Mentored team members on version control best practices and collaborative coding standards, significantly improving team efficiency and reducing code integration issues.
- Led the agile development cycle

PROJECTS

GoReactChat – Real-Time Chat Application

09 Feb, 2025 - 19 Mar, 2025

Key Skills:

- Go
- Gorilla Mux
- Gorilla Mux
- WebSockets
- JSON
- JWT Auth
- ReactJS
- Tailwind CSS
- PostgreSQL
- MongoDB
- Redis

Project Link: <https://github.com/Kalpesh-Vala/go-react-chat>

- Designed and developed a full-stack real-time chat platform with WebSocket-based live messaging and secure user authentication via JWT.
- Built a scalable backend in Go, using Gorilla Mux for routing and WebSocket handlers for real-time communication.
- Implemented multi-database architecture with PostgreSQL for user data, MongoDB for message storage, and Redis for caching and real-time user status tracking.
- Integrated AI chatbot for personal conversations and enabled both private and group chat functionality.
- Added advanced messaging features including message reactions and user-specific message services.
- Developed a modern frontend using ReactJS with Vite bundler and Tailwind CSS for a fast, responsive, and dynamic user interface.
- Engineered a clean, modular codebase ensuring maintainability, performance, and future extensibility.

MyShops - Backend System

08 Jun, 2024 - 02 Jul, 2024

Team Size: 1

- Key Skills:
- Java
- Spring Boot
- REST API
- JPA
- Maven
- Docker

Project Link: <https://github.com/Kalpesh-Vala/my-shops>

- Developed a scalable e-commerce backend with core modules like product, user, cart, and order management.
- Implemented layered architecture (Controller, Service, Repository) with custom exception handling for robustness.
- Utilized JPA with Hibernate for efficient database operations and Docker for portable deployment and testing.
- Organized the project using Maven for modular and production-ready development.

Java E-Commerce Web Application

Key Skills:

- Java
- JDBC
- MySQL
- MVC Architecture
- Gradle Build Tool
- HTML/CSS (JSP Styling)
- Session Management
- Apache Tomcat Server

Project Link: <https://github.com/Kalpesh-Vala/JAVA-E-commerce>

- Built a fully functional e-commerce web application with customer login, registration, product listings, cart, and profile features.
- Followed MVC architecture using Servlets and JSP; implemented secure database operations via JDBC.
- Designed responsive JSP interfaces styled with HTML/CSS and integrated session handling and form validation.
- Structured backend logic using modular DAO and Entity layers for clean database interaction.

PUBLICATIONS / RESEARCH / WHITE PAPERS

Steganography Research: Techniques, Findings & Future Directions

Research (Poster Presentation) | Mentor: Dr. Lata Gohil | No. of Authors: 2

- Key Skills:
- Cryptography
- Steganography
- Quantum Computing
- Image Processing

Awarded Second Position (jointly) in the Poster Presentation at the National Research Symposium 2025, themed "Recent Trends in Engineering", organized by the Institute of Technology, Nirma University. The presented work, titled "Steganography Research: Techniques, Findings & Future Directions", was recognized for its originality, research depth, and clarity of presentation. The research involved an analytical study of existing steganographic techniques, identification of gaps in current methods, and a proposed roadmap for future advancements in secure data embedding. The project is currently in the implementation phase, with documentation and design already completed.

ASSESSMENTS / CERTIFICATIONS

Complete Kubernetes Guide (Udemy)

- Key Skills:
- Docker
- Kubernetes
- Microservices

HackNuThon 6.0

- Key Skills:
- Python
- AIML
- Testing

AWS Academy Graduate - AWS Academy Cloud Foundations

**Key Skills:** AWS

## SEMINARS / TRAININGS / WORKSHOPS

**Aspire Leaders Program - Module 1: Personal and Professional Development** Institute Name: Aspire Institute Inc.

**Key Skills:** Personal Development

**Aspire Leaders Program Module 2: World-Class, Educator led Masterclasses**

Institute Name: Aspire Institute Inc.

**Key Skills:** Communication Skills

## CO-CURRICULAR ACTIVITIES

- o Guest Speaker – Vidyadhan Foundation Student Camps (2020 – Present) Regularly invited by Vidyadhan Foundation to speak at their annual student camps, guiding engineering aspirants on career planning, exam preparation, and personal development.

## WEB LINKS

- o Personal - <https://kalpesh-vala.me>